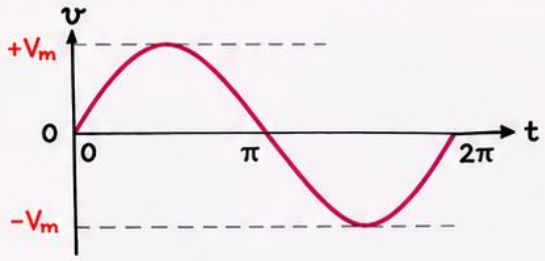
Different Types of AC Waveforms

An AC (Alternating Current) waveform is a voltage or current that varies with time and changes its magnitude and direction periodically.

1 Sine Wave (Sinusoidal Waveform)



- Most common AC waveform.
- Mathematical form:

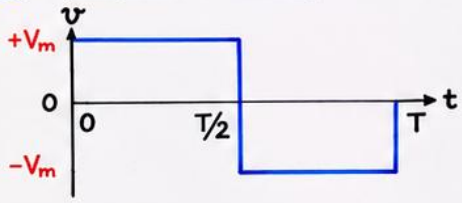
$$v(t) = V_m \sin \omega t$$

- Smooth and continuous.
- Used in power systems, transformers, motors.

Key Features
- ✓ Peak value = V_m
- ✓ Frequency = f
- ✓ Time period = $T = \frac{1}{f}$
- ✓ Symmetrical about zero

Applications
- Power generation
- AC circuits
- Communication signals

2 Square Wave

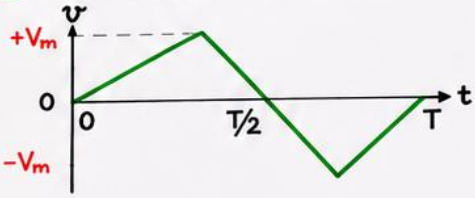


- Alternates between $+V_m$ and $-V_m$ instantly.
- Has sharp transitions.
- Contains odd harmonics.
- Used in digital circuits and inverters.

Key Features
- ✓ Sudden changes (discontinuities)
- ✓ Easy to generate

Applications
- Digital electronics
- Switching circuits
- Gate drive signals

3 Triangular Wave

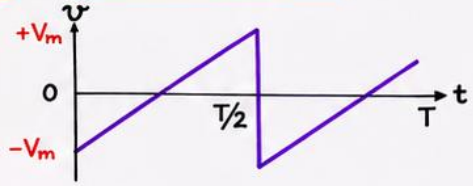


- Linear rise and fall between $+V_m$ and $-V_m$.
- No sudden jumps.
- Contains both odd and even harmonics.
- Used in signal generation, ramp circuits.

Key Features
- ✓ Linear segments
- ✓ Symmetrical about zero

Applications
- Function generators
- Modulation circuits

4 Sawtooth Wave

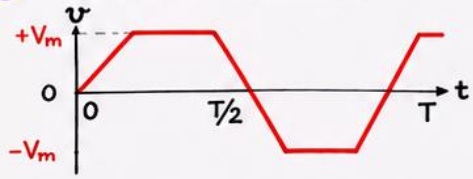


- Linear rise (or fall) followed by sudden drop (or rise).
- Not symmetrical about zero.
- Contains both odd and even harmonics.
- Used in oscillators, sweep generators.

Key Features
- ✓ Sudden drop at end of period
- ✓ Used for time base

Applications
- CRO time base
- Oscillators, Scanners

5 Trapezoidal Wave



- Combination of square and triangular wave.
- Has both flat and sloped portions.
- Less distortion than square wave.
- Used in motor drives, UPS etc.

Key Features
- ✓ Nearly symmetrical
- ✓ Less harmonic distortion

Applications
- Power electronics
- Modulation, Control systems

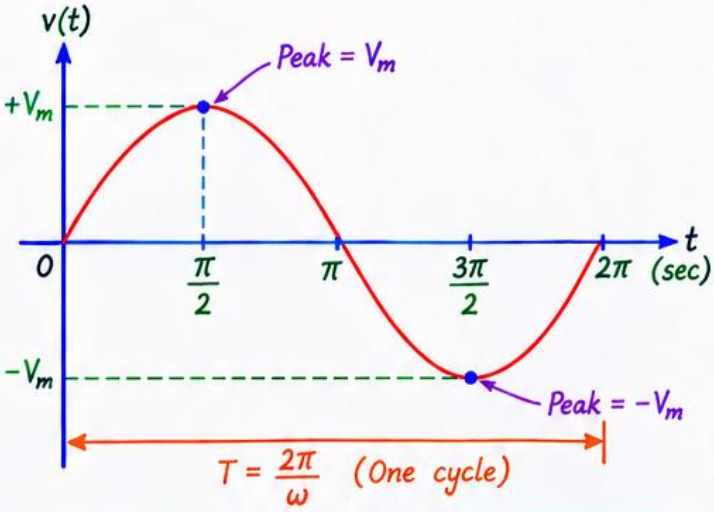
Note: Sine wave is the ideal waveform for AC power systems because it has minimum harmonic content and is smooth in nature.

STEADY STATE AC CIRCUITS

Representation of sinusoidal waveforms

→ In steady state AC analysis, voltages and currents vary sinusoidally with time.

→ A sinusoidal waveform is completely described by its amplitude, frequency (or angular frequency) and phase angle.



General equation :

$$v(t) = V_m \sin(\omega t + \phi)$$

where,

- V_m = Maximum (peak) value (V or A)
- ω = Angular frequency (rad/s)
- ϕ = Phase angle (rad or degree)
- T = Time period (s) = $\frac{2\pi}{\omega}$
- f = Frequency (Hz) = $\frac{1}{T} = \frac{\omega}{2\pi}$

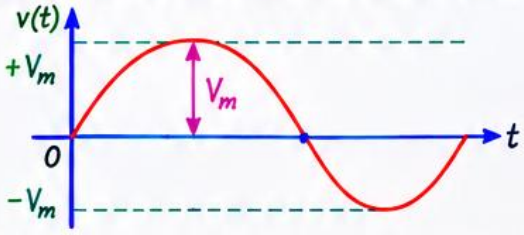
Important features :

- At $t = 0$, $v(0) = V_m \sin\phi$
- Positive half cycle : 0 to π
- Negative half cycle : π to 2π
- One complete cycle : 0 to 2π (or T sec)

---

1. Amplitude (V_m) :

→ Maximum value of the waveform measured from the zero reference line.



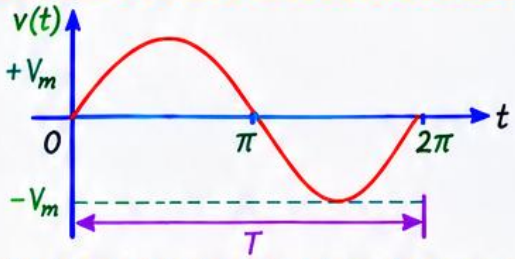
Peak to peak value :

$$V_{pp} = 2V_m$$

---

2. Time period (T) :

→ Time taken by the waveform to complete one cycle.



$$T = \frac{2\pi}{\omega} = \frac{1}{f} \text{ (seconds)}$$

---

3. Frequency (f) :

→ Number of cycles per second.

$$f = \frac{1}{T} = \frac{\omega}{2\pi} \text{ Hz}$$

Common frequencies :

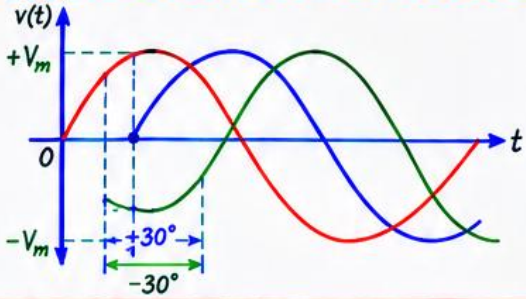
- Power frequency = 50 Hz (India) or 60 Hz (USA)

---

4. Phase angle (ϕ) :

→ Represents the initial angular displacement of the waveform at $t = 0$.

→ Determines lead or lag of the waveform.



Phase shift examples :

- $\phi = 0°$ (no phase shift)
- $\phi = +30°$ (leads)
- $\phi = -30°$ (lags)

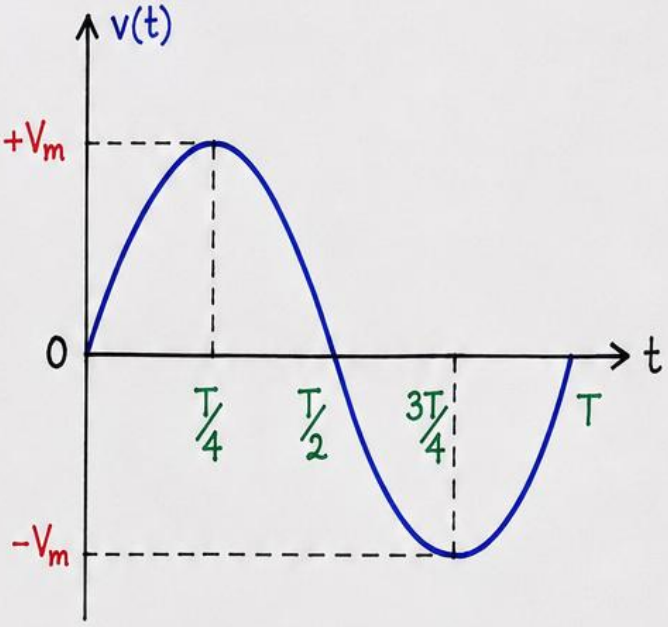
★ Sinusoidal representation is the most convenient form for analysis of steady state AC circuits using phasors and network theorems.

Phasor Representation of AC waveform

1. AC Waveform in Time Domain

A sinusoidal alternating quantity can be represented in time domain as:

$$v(t) = V_m \sin(\omega t + \theta)$$

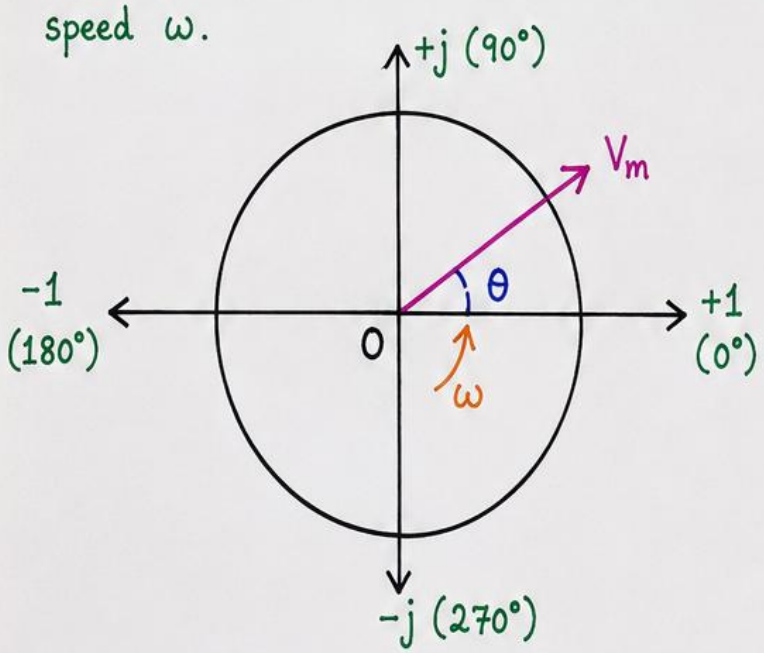


where,

- V_m = Peak (maximum) value
- ω = Angular frequency = $2\pi f$ rad/s
- θ = Initial phase angle (in radians)

2. Phasor Representation

A sinusoidal quantity can be represented by a rotating vector (phasor) of constant magnitude V_m rotating in anticlockwise direction with angular speed ω.



Phasor at any time t

- Length = V_m (constant)
- Angle with +ve real axis = $\omega t + \theta$
- Projection on vertical axis gives instantaneous value v(t) = $V_m \sin(\omega t + \theta)$

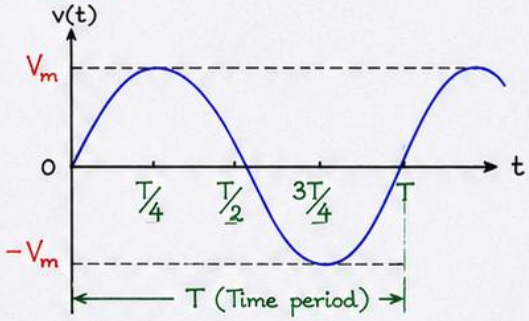
Different ways of representation of Sinusoidal waveforms

A sinusoidal waveform can be represented in different forms to study its magnitude, phase, frequency and relationships easily.

The commonly used representations are :

1. Time domain (Waveform) Representation

It is the most common representation in which the instantaneous value of the quantity is plotted with respect to time.



Equation :

$v(t) = V_m \sin \omega t$

where,

V_m = Peak value

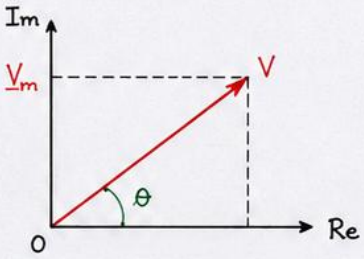
T = Time period

$\omega = 2\pi f$ rad/s

f = Frequency (Hz)

2. Phasor (Vector) Representation

A sinusoidal quantity is represented by a rotating vector (phasor) of magnitude equal to its RMS value, rotating at angular speed ω.



Phasor :

$\vec{V} = V \angle \theta$

where,

V = RMS value

θ = Phase angle (rad or °)

3. Analytical (Mathematical) Representation

The sinusoidal waveform is represented by a mathematical equation.

Equation :

$v(t) = V_m \sin(\omega t + \theta)$

where,

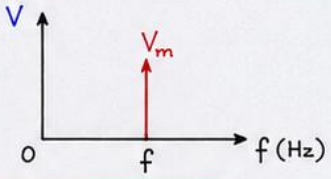
V_m = Peak value

$\omega = 2\pi f$

θ = Phase angle

4. Frequency domain (Spectral) Representation

A sinusoidal waveform of frequency f is represented by a single line (spike) at that frequency on a frequency axis.



Shows the magnitude of the sinusoid at its frequency f.

5. Complex (Exponential) Representation

A sinusoidal quantity is represented using complex exponentials (Euler's formula).

Equation :

$v(t) = \text{Re}\{\vec{V} e^{j\omega t}\}$

where, $\vec{V} = V \angle \theta$

$j = \sqrt{-1}$

Very useful in AC circuit analysis and signal processing.

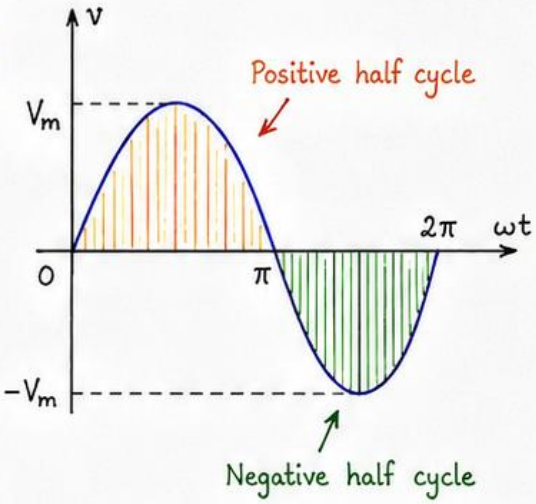
Summary :

1. Time domain → actual waveform v/s time.
2. Phasor domain → rotating vector of RMS value.
3. Analytical → mathematical equation.
4. Frequency domain → single line at frequency f.
5. Complex form → using complex exponentials.

Note : All these representations are equivalent and used as per the requirement of the analysis.

Understanding of Average Value of AC Waveform and Its Derivation

* Average value (or Mean value) of an alternating quantity is defined as the arithmetic mean of the instantaneous values over one half cycle.



Why over half cycle ?

→ In a complete cycle, the positive and negative areas are equal and cancel each other.

→ Hence, average over one complete cycle is zero.

1. Average value of a sinusoidal waveform (over half cycle)

Let, $v = V_m \sin \omega t$ be a sinusoidal voltage.

Its average value over one half cycle (0 to π) is given by,

$$V_{AV} = \frac{1}{\pi} \int_0^{\pi} v \, d(\omega t) \qquad \text{--- (1)}$$

Substituting $v = V_m \sin \omega t$,

$$V_{AV} = \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t \, d(\omega t)$$

$$= \frac{V_m}{\pi} \int_0^{\pi} \sin \omega t \, d(\omega t)$$

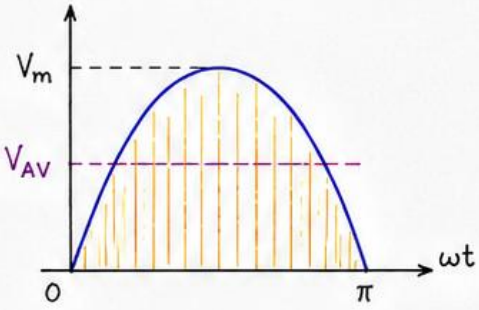
$$= \frac{V_m}{\pi} \left[-\cos \omega t \right]_0^{\pi} \qquad (\because \int \sin x \, dx = -\cos x)$$

$$= \frac{V_m}{\pi} \left[-\cos \pi + \cos 0 \right]$$

$$= \frac{V_m}{\pi} \left[-(-1) + 1 \right] = \frac{2V_m}{\pi}$$

$$V_{AV} = \frac{2V_m}{\pi} = 0.637 \, V_m \qquad \text{--- (2)}$$

Explanation (Graphical)



The average value (V_{AV}) is equal to the area under the curve over half cycle (0 to π) divided by the base (π).

2. Average value over full cycle (0 to 2π)

Over a full cycle, the positive and negative areas are equal.

$$V_{AV} = \frac{1}{2\pi} \int_0^{2\pi} V_m \sin \omega t \, d(\omega t) = 0$$

(Because the waveform is symmetrical)

3. Average value of current

If $i = I_m \sin \omega t$, then

$$I_{AV} = \frac{2I_m}{\pi} = 0.637 \, I_m$$

(over one half cycle)

Key Points :

* Average value of a sine wave over half cycle = $\frac{2}{\pi}$ times the maximum value ($\approx 0.637 \, V_m$).
* Average value over a full cycle = 0.
* It is also called Mean (or Arithmetic mean) value.

Understanding of RMS Value of AC Waveform and Its Derivation

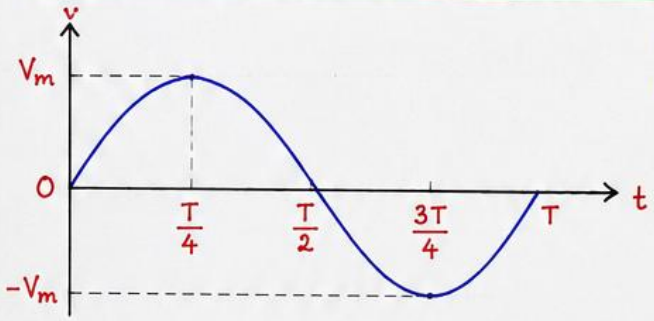
1. What Is RMS Value?

The RMS (Root Mean Square) value of an AC waveform is the equivalent DC value which would produce the same power (or heating effect) in a resistor as the AC waveform does over one complete cycle.

2. Why RMS Value Is Important?

- It represents the effective value of an AC quantity.
- Used in measuring AC voltages, currents, and powers.
- For a sine wave, RMS value is related to the peak (maximum) value by a constant factor.

3. Derivation of RMS Value For A Sinusoidal Waveform



Let $v(t) = V_m \sin \omega t$ be an AC voltage waveform.

V_m = Peak (maximum) value
T = Time period
ω = Angular frequency = $2\pi f$
f = Frequency

Definition: The RMS value (V_{rms}) is given by

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

It means the square root of the mean of the squared instantaneous values over one period.

Derivation:

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} = \sqrt{\frac{1}{T}\int_0^T (V_m \sin \omega t)^2\,dt}$$

$$= \sqrt{\frac{V_m^2}{T}\int_0^T \sin^2 \omega t\,dt} \qquad [\because V_m \text{ is constant}]$$

We know that $\sin^2 \omega t = \frac{1 - \cos 2\omega t}{2}$

$$\therefore V_{rms} = \sqrt{\frac{V_m^2}{T}\int_0^T \frac{1 - \cos 2\omega t}{2}\,dt} = \sqrt{\frac{V_m^2}{2T}\left[\int_0^T 1\,dt - \int_0^T \cos 2\omega t\,dt\right]}$$

$$= \sqrt{\frac{V_m^2}{2T}\left[T - 0\right]} \qquad \because \int \cos 2\omega t\,dt = \frac{\sin 2\omega t}{2\omega} \text{ from 0 to T}$$

$$= \frac{\sin 2\omega T - \sin 0}{2\omega} = 0$$

$$= \sqrt{\frac{V_m^2}{2}} = \frac{V_m}{\sqrt{2}}$$

$$\therefore V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707\,V_m \quad \text{(For sine wave)}$$

Results & Special Cases:

- For a sinusoidal voltage: $V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707\,V_m$
- For a sinusoidal current: $I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707\,I_m$
- Peak value = $\sqrt{2}$ × RMS value (i.e., $V_m = \sqrt{2}\,V_{rms}$)

Example:

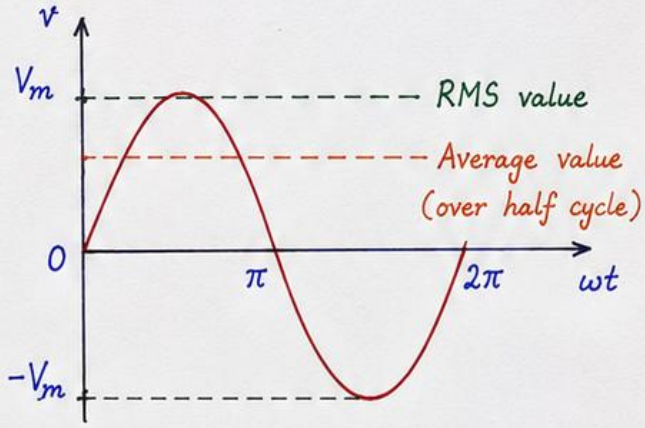
If V_{rms} = 230 V (AC mains), then peak value

$V_m = \sqrt{2} \times 230 = 325$ V (approx.)

Form Factor and Peak Factor

Introduction :

In an alternating quantity, the amplitude changes continuously with time. To compare the value of AC quantities, certain factors are used.
Two important factors are form factor and peak (crest) factor.



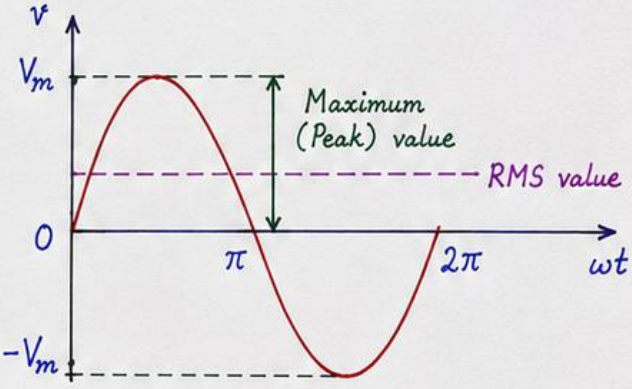
1. Form Factor :

The form factor of an AC quantity is defined as the ratio of RMS value to the average (mean) value of the quantity over one half cycle.

Formula :

$$\text{Form Factor } (K_f) = \frac{\text{RMS value}}{\text{Average value}}$$

Note : Average value is taken over one half cycle because the average over a full cycle is zero.



2. Peak (Crest) Factor :

The peak or crest factor of an AC quantity is defined as the ratio of maximum (peak) value to the RMS value.

Formula :

$$\text{Peak Factor } (K_p) = \frac{\text{Maximum (Peak) value}}{\text{RMS value}}$$

Values for a Pure Sinusoidal Wave :

For a sinusoidal wave,

RMS value, $V_{rms} = \frac{V_m}{\sqrt{2}}$

Average value, $V_{avg} = \frac{2V_m}{\pi}$

Therefore,

Form Factor $K_f = \frac{V_{ms}/\sqrt{2}}{V_{rms}/\pi} = \frac{\pi}{2\sqrt{2}} = 1.11$

Peak Factor $K_p = \frac{V_m}{V_m/\sqrt{2}} = \sqrt{2} = 1.414$

Summary :

* Form Factor $(K_f) = \frac{\text{RMS value}}{\text{Average value (over half cycle)}}$ — For sine wave, $K_f = 1.11$
* Peak Factor $(K_p) = \frac{\text{Maximum (Peak) value}}{\text{RMS value}}$ — For sine wave, $K_p = 1.414$

Understanding of Lagging and Leading in Sinusoidal Waveforms

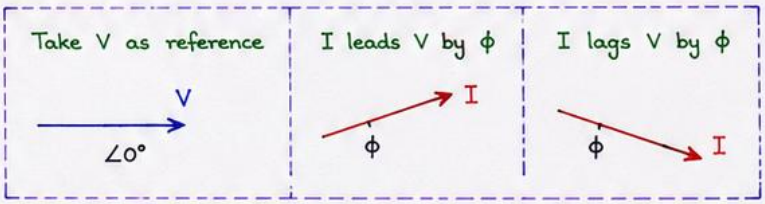
1. Introduction :

In AC circuits, voltage and current are sinusoidal quantities. They may not reach their zero, positive or negative peaks at the same instant. This time difference is called phase difference (ϕ).

ω = Angular frequency (rad/s)
T = Time period = $2\pi/\omega$ (s)
ϕ = Phase difference (radians or degrees)

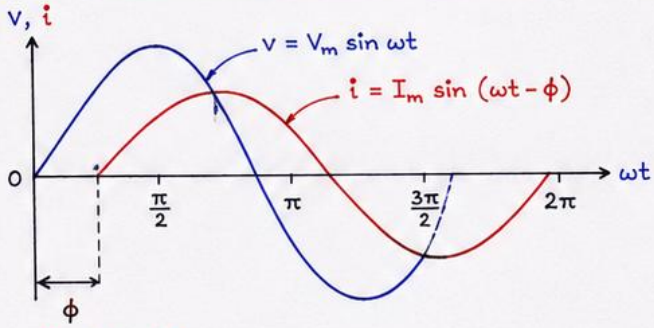
2. Reference :

In AC circuit analysis, voltage is usually taken as the reference phasor.



3. Lagging and Leading Waveforms :

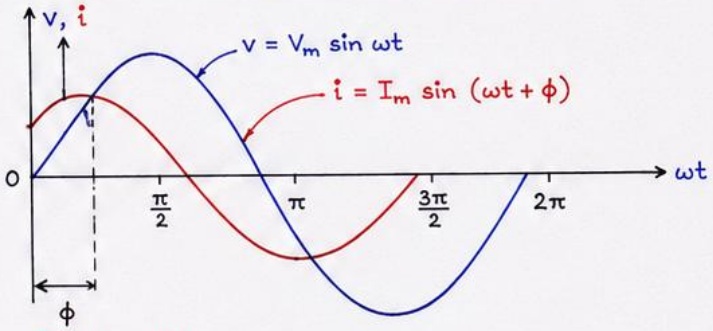
(a) Current Lags voltage (I lags V)



Time by which current lags

Here, current reaches any point ϕ radians (or ϕ/ω seconds) later than voltage.

(b) Current Leads voltage (I leads V)



Time by which current leads

Here, current reaches any point ϕ radians (or ϕ/ω seconds) earlier than voltage.

4. Phase Relationship (Mathematical Form)

If current lags voltage by ϕ,

$$v = V_m \sin \omega t$$
$$i = I_m \sin (\omega t - \phi)$$

If current leads voltage by ϕ,

$$v = V_m \sin \omega t$$
$$i = I_m \sin (\omega t + \phi)$$

Positive ϕ means current leads.

Negative ϕ means current lags.

5. Key Points :

- When current lags voltage, power factor is lagging (inductive circuit).
- When current leads voltage, power factor is leading (capacitive circuit).
- Phase difference ϕ can be expressed in radians or degrees.
- 1 radian = 57.3°, $1° = \frac{\pi}{180}$ rad.
- Time difference, $\Delta t = \frac{\phi}{\omega}$ seconds.

Example :

If $\phi = 30°$, $\omega = 100$ rad/s

Time difference, $\Delta t = \frac{\phi}{\omega} = \frac{30° \times \frac{\pi}{180}}{100}$

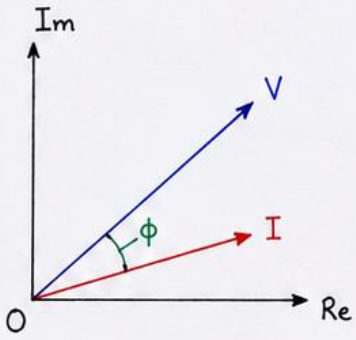
$= \frac{\pi}{600}$ s = 5.24 ms

Understanding of Lagging and Leading Using Phasor Representation

Phasor representation is a graphical method used to represent sinusoidal quantities in terms of magnitude and phase angle.

1. Lagging (Current lags Voltage)

When current lags behind voltage by an angle ϕ, the circuit is said to be lagging.



Here,
V = Voltage phasor
I = Current phasor
ϕ = Phase angle (V leads I)

Example :

- RL Circuit (Inductive circuit)
- RLC circuit when $X_L > X_C$
- Power factor is lagging

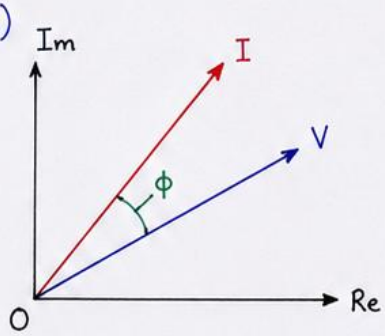
Key Points :

- Current is behind Voltage.
- ϕ is positive.
- Power factor = cos ϕ (lagging)

---

2. Leading (Current leads Voltage)

When current leads voltage by an angle ϕ, the circuit is said to be leading.



Here,
V = Voltage phasor
I = Current phasor
ϕ = Phase angle (I leads V)

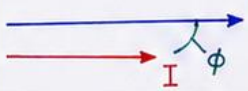
Example :

- RC Circuit (Capacitive circuit)
- RLC circuit when $X_C > X_L$
- Power factor is leading

Key Points :

- Current is ahead of Voltage.
- ϕ is negative.
- Power factor = cos ϕ (leading)

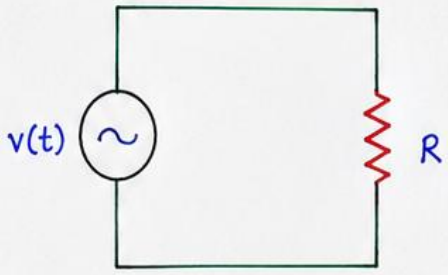
Summary

Type	Phasor Relation	Phase Angle (ϕ)	Power Factor	Circuit Example
Lagging (I lags V)		Positive ($0° < \phi < 90°$)	cos ϕ (lagging)	RL circuit (Inductive)
Leading (I leads V)		Negative ($-90° < \phi < 0°$)	cos ϕ (leading)	RC circuit (Capacitive)

Note: V leads I in lagging circuits, I leads V in leading circuits.

Analysis of single-phase AC circuits consisting of R only

1. Circuit Diagram :



2. Applied Voltage :

$$v(t) = V_m \sin \omega t$$

where,

V_m = Peak value of voltage (Volts)

ω = Angular frequency ($= 2\pi f$ rad/s)

f = Frequency (Hz)

3. Current through R :

By Ohm's law,

$$i(t) = \frac{v(t)}{R} = \frac{V_m \sin \omega t}{R}$$

$$i(t) = I_m \sin \omega t$$

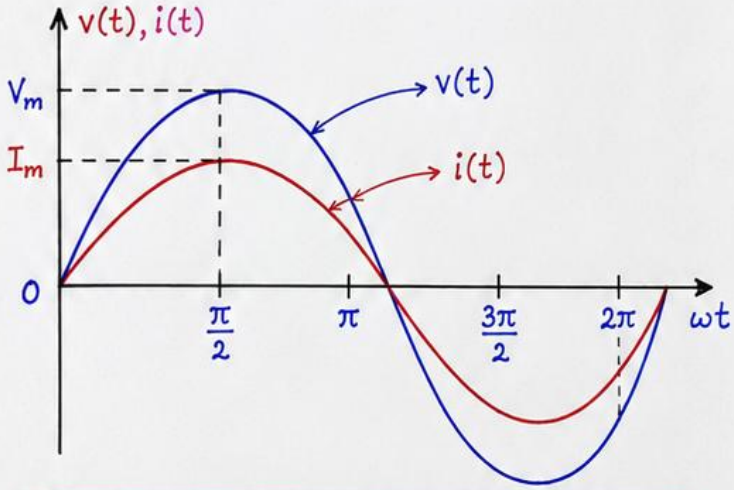
where,

$$I_m = \frac{V_m}{R} = \text{Peak value of current (Amps)}$$

Also,

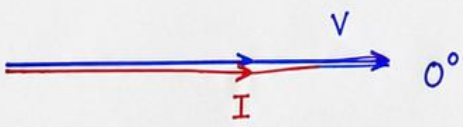
$$V_{rms} = \frac{V_m}{\sqrt{2}} \quad , \quad I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{V_{rms}}{R}$$

4. Voltage and Current Waveforms :



In a pure resistive AC circuit, voltage and current are in phase with each other.

5. Phasor Diagram :



V and I are in phase ($\phi = 0°$)

6. Power in R only circuit :

Instantaneous power,

$$p(t) = v(t)\, i(t) = V_m \sin \omega t \,.\, I_m \sin \omega t$$

$$= \frac{V_m I_m}{2} \left(1 - \cos 2\omega t \right)$$

Average (Real) power,

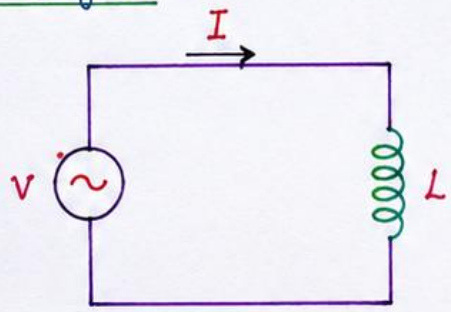
$$P = V_{rms}\, I_{rms} = \frac{V_{rms}^2}{R} = I_{rms}^2 R \quad \text{(Watts)}$$

7. Key Points :

* Current is directly proportional to voltage. ($i \propto v$)
* Voltage and current are in phase. ($\phi = 0°$)
* Power factor = $\cos \phi = \cos 0° = 1$ (Unity)
* All the power is real power.

Analysis of single-phase AC circuits consisting of L only

1) Circuit Diagram :



Consider a pure inductor L connected to a single-phase AC supply of voltage v.

* There is no resistance in the circuit.
* Hence, current flows only due to the inductance.

2) Voltage and Current Equations :

Let the supply voltage be

$$v = V_m \sin \omega t$$

For an inductor,

$$\boxed{v = L \frac{di}{dt}}$$

$$\frac{di}{dt} = \frac{V_m}{L} \sin \omega t$$

Integrating,

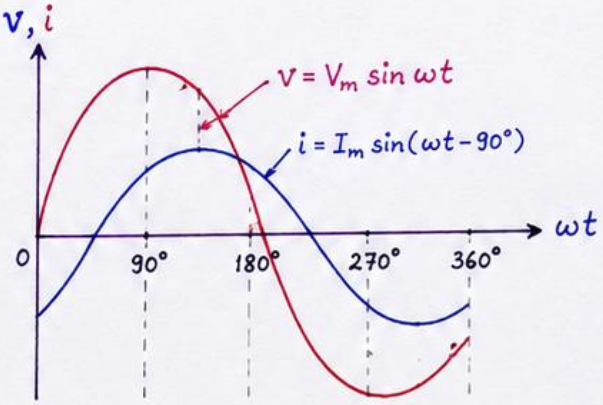
$$\boxed{i = -\frac{V_m}{\omega L} \cos \omega t + C}$$

At $t = 0$, $i = 0 \Rightarrow C = 0$

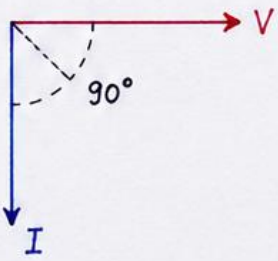
$$\boxed{i = \frac{V_m}{\omega L} \sin(\omega t - 90°)}$$

Thus, in a pure inductor, current lags the voltage by 90°.

3) Waveforms :



4) Phasor Diagram :



Current (I) lags voltage (V) by 90° in a pure inductor.

5) Impedance of Inductor :

Impedance, $Z = \frac{V}{I} = \frac{V_m}{I_m} \angle 90°$

$$\boxed{Z_L = j\omega L} \quad \text{(Inductive Reactance)}$$

where, $X_L = \omega L$ (in ohms)

6) Key Points :

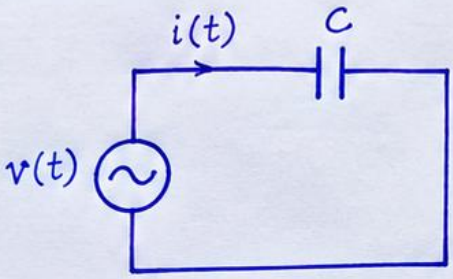
* In a pure inductor, current lags voltage by 90°.
* Impedance is purely imaginary and equal to $j\omega L$.
* Average power consumed is zero.
* Only reactive power is exchanged between source and inductor.

Summary : $v = V_m \sin \omega t$, $i = I_m \sin(\omega t - 90°)$, $Z_L = j\omega L$, I lags V by 90°

Analysis of single-phase AC circuits consisting of C only

1. Pure Capacitive AC Circuit:

Consider a single-phase AC circuit consisting of a capacitor C connected to an AC source.



Source voltage:

$$v(t) = V_m \sin \omega t$$
$$= V_m \angle 0^\circ$$

Capacitive reactance:

$$X_C = \frac{1}{\omega C}$$

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2. Current in Capacitive Circuit:

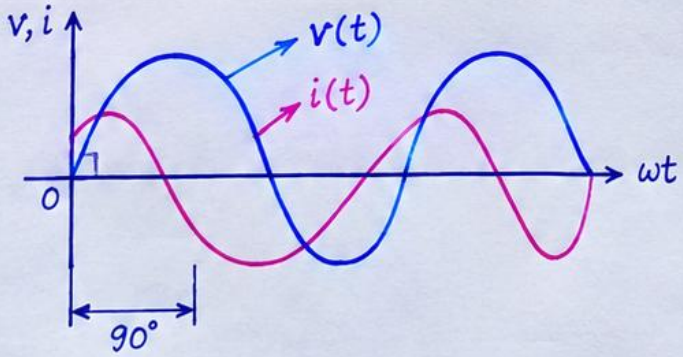
For a pure capacitor,

$$i(t) = C\left(\frac{dv}{dt}\right) = C V_m \omega \cos \omega t$$
$$= I_m \sin(\omega t + 90^\circ)$$

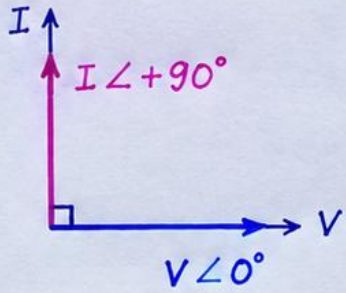
where $I_m = \omega C V_m = \frac{V_m}{X_C}$

Key result:
In a pure capacitive circuit, the current leads the voltage by 90°.
(i.e., current leads voltage)

3. Waveforms:



4. Phasor Diagram:



5. Power:

Instantaneous power:

$$p(t) = v(t)i(t) = V_m I_m \sin\omega t \cdot \sin(\omega t + 90^\circ)$$
$$= \frac{V_m I_m}{2} \sin 2\omega t$$

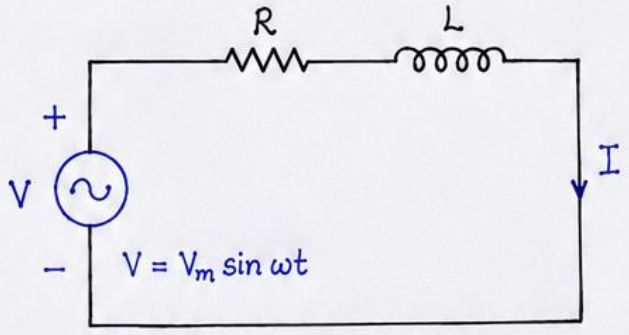
Average power:

$$P = VI \cos\phi = 0$$

(No real power is consumed in a pure capacitive circuit)

Analysis of Single-phase AC Circuits Consisting of Series of R & L

1. Circuit Diagram :



2. Voltage Equation :

By KVL,

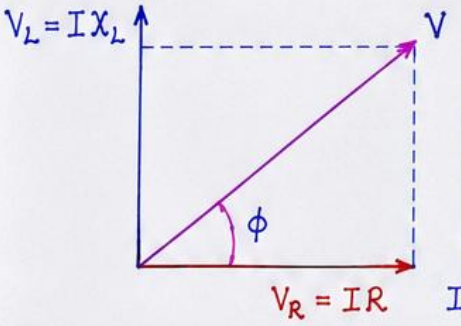
$$V = V_R + V_L$$

where,

$$V_R = iR \quad , \quad V_L = L\frac{di}{dt}$$

$$\therefore V = iR + L\frac{di}{dt} \quad \text{— (1)}$$

3. Phasor Diagram :



V leads V_R by ϕ and V_L leads I by 90°.

where, $X_L = \omega L$ (Inductive reactance)

4. Power in R-L Series Circuit :

* Real Power (P) = $I^2R = VI\cos\phi$ (watts)
* Reactive Power (Q) = $I^2X_L = VI\sin\phi$ (VAR)
* Apparent Power (S) = VI (VA)
* Power Factor ($\cos\phi$) = $\frac{R}{Z}$ (lagging)

5. Impedance :

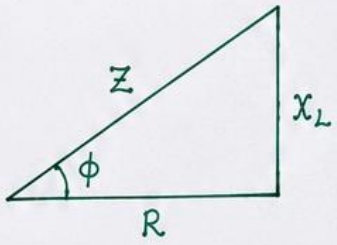
Inductive reactance, $X_L = \omega L$

Impedance, $Z = R + jX_L$

$$|Z| = \sqrt{R^2 + X_L^2} \quad \text{(ohms)}$$

$$\phi = \tan^{-1}\left(\frac{X_L}{R}\right)$$

$$\boxed{Z = |Z| \angle \phi}$$ (Impedance triangle)



6. Current :

$$I = \frac{V}{Z} \quad \text{(phasor)}$$

$$\boxed{I = \frac{V}{|Z|} \angle -\phi}$$

⇒ Current lags the supply voltage by ϕ (lagging power factor)

7. Voltage Drops :

$V_R = IR$ (in phase with I)

$V_L = IX_L$ (leads I by 90°)

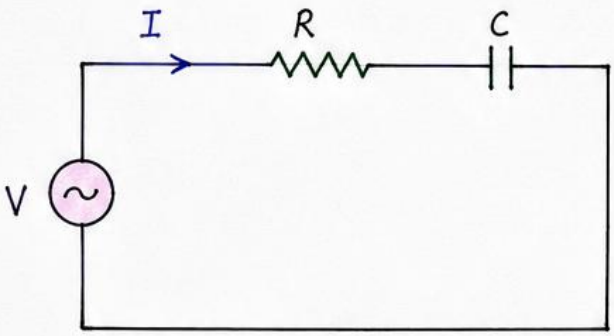
8. Summary :

* In a series R-L circuit, current lags the supply voltage by angle ϕ.
* Voltage across R is in phase with current.
* Voltage across L leads the current by 90°.
* Power factor is lagging.

Conclusion : The R-L series circuit has impedance (Z) with a lagging power factor. Real power is consumed in R while L stores energy in the form of magnetic field.

Analysis of Single-phase AC Circuits Consisting of Series of R & C

1. Circuit Diagram :



2. Impedance of the circuit :

$$X_C = \frac{1}{\omega C} \quad (\Omega)$$

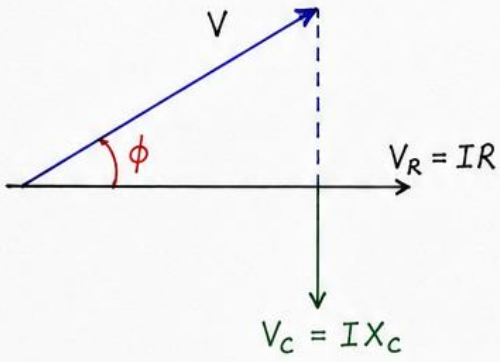
$$Z = R - jX_C = R - j\frac{1}{\omega C} \quad (\Omega)$$

$$|Z| = \sqrt{R^2 + X_C^2} \quad (\Omega)$$

$$\phi = \tan^{-1}\left(\frac{X_C}{R}\right)$$

(Current leads the voltage)

3. Phasor Diagram :



In series R-C circuit :

- Current (I) leads the supply voltage (V) by angle ϕ.
- V, V_R and V_C form a right-angled triangle.
- V_C lags behind I by 90°.

4. Voltage Relations :

$$V_R = IR$$

$$V_C = IX_C = \frac{I}{\omega C}$$

$$V = \sqrt{V_R^2 + V_C^2}$$

$$\tan\phi = \frac{V_C}{V_R} = \frac{X_C}{R}$$

5. Current in the circuit :

$$I = \frac{V}{Z} = \frac{V}{R - jX_C}$$

$$|I| = \frac{V}{|Z|} = \frac{V}{\sqrt{R^2 + X_C^2}}$$

(Current leads V by angle ϕ)

6. Power in the circuit :

- Real Power, $P = I^2R = \frac{V^2R}{R^2 + X_C^2}$ watts
- Reactive Power, $Q = -I^2X_C = -\frac{V^2X_C}{R^2 + X_C^2}$ VAR (negative)
- Apparent Power, $S = VI = \frac{V^2}{\sqrt{R^2 + X_C^2}}$ VA

Power Factor :

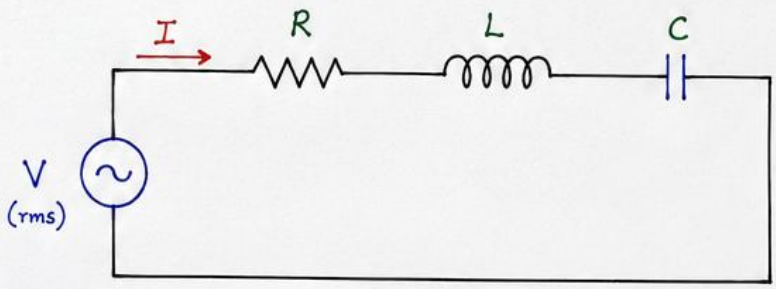
$$\cos\phi = \frac{R}{|Z|} = \frac{R}{\sqrt{R^2 + X_C^2}}$$

(leading)

Summary : In a series R-C circuit, the current leads the supply voltage by angle ϕ. The circuit has negative reactive power (capacitive nature) and leading power factor.

Analysis of single-phase AC circuits consisting of series of R, L & C

A single-phase AC circuit consisting of resistance (R), inductance (L) and capacitance (C) connected in series is considered.



R = Resistance (ohms)
L = Inductance (henry)
C = Capacitance (farad)
V = Supply Voltage (rms)
I = Circuit Current (rms)

$\omega = 2\pi f$ (rad/s)
f = frequency (Hz)

1) IMPEDANCES :

Resistance, $Z_R = R$ (in phase with I)

Inductive reactance, $Z_L = jX_L = j\omega L$ (I lags V by 90°)

Capacitive reactance, $Z_C = -jX_C = -\frac{j}{\omega C}$ (I leads V by 90°)

Net Reactance :
$X = X_L - X_C$
(positive → inductive)
(negative → capacitive)

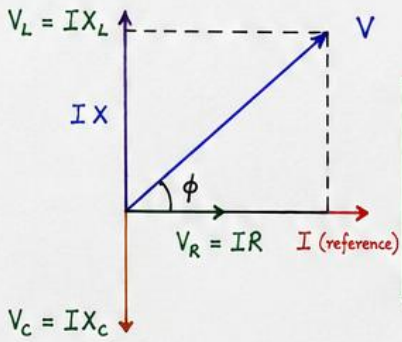
2) TOTAL IMPEDANCE (Z) :

In a series R-L-C circuit,

$Z = R + j(X_L - X_C) = R + jX$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$$
$$Z = |Z| \angle \phi$$

3) PHASOR DIAGRAM (Voltage Triangle) :



$V = IZ$
$V = \sqrt{V_R^2 + (V_L - V_C)^2}$
$\tan\phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$

4) CIRCUIT CURRENT (I) :

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}}$$

If $X_L > X_C$ → Circuit is inductive (I lags V by ϕ)
If $X_L < X_C$ → Circuit is capacitive (I leads V by ϕ)
If $X_L = X_C$ → Circuit is at resonance, $\phi = 0°$

5) VOLTAGE DROPS :

$V_R = IR$ (in phase with I)
$V_L = IX_L$ (leads I by 90°)
$V_C = IX_C$ (lags I by 90°)

Supply voltage, $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

6) POWER :

Real Power, $P = I^2R = VI\cos\phi$ (watts)
Reactive Power, $Q = I^2(X_L - X_C) = VI\sin\phi$ (VAR)
Apparent Power, $S = VI = I^2Z$ (VA)
Power Factor, $\cos\phi = \frac{P}{S}$

At resonance ($X_L = X_C$): $X = 0$, $Z = R$ (minimum), $I = \frac{V}{R}$ (maximum), $\phi = 0°$, Power factor = 1

Real Power, Reactive Power, Apparent Power, Power Factor

In AC circuits, power is not constant as in DC circuits. It varies sinusoidally. Therefore, we use average or RMS values to define different types of power.

① Real Power (P)

The average power actually consumed by the resistive part of the load and converted into useful work.

$$P = V I \cos\phi \quad \text{(Watt)}$$

where, V = RMS voltage (V)
I = RMS current (A)
ϕ = Phase angle between V and I

Unit : Watt (W)

- It is positive for both inductive and capacitive circuits.
- It represents the actual power used in doing useful work.

② Reactive Power (Q)

The power which oscillates between the source and reactive elements (L or C) of the circuit. It does no useful work.

$$Q = V I \sin\phi \quad \text{(var)}$$

- Q is positive for inductive load (lagging current).
- Q is negative for capacitive load (leading current).

Unit : Var (Volt-Ampere Reactive)

- It indicates the amount of power that oscillates between source and reactive elements.
- It is necessary for the working of inductors and capacitors.

③ Apparent Power (S)

The product of RMS voltage and RMS current. It is the total power supplied by the source.

$$S = V I \quad \text{(VA)}$$

Also, $S^2 = P^2 + Q^2$ or $S = \sqrt{P^2 + Q^2}$

Unit : VA (Volt-Ampere)

- It includes both real and reactive powers.
- It is always greater than or equal to real power. $(S \geq P)$

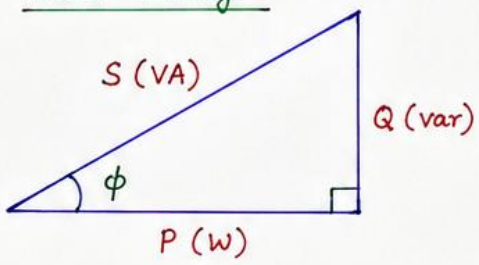
④ Power Factor ($\cos\phi$)

It is the ratio of real power to apparent power.

$$\cos\phi = \frac{P}{S} = \frac{P}{VI}$$

- $0 < \cos\phi < 1$ for practical circuits.
- $\cos\phi = 1$ for purely resistive load.
- Low power factor means more current for the same power.
- Higher power factor is desirable.

Power Triangle



$$\cos\phi = \frac{\text{adjacent side}}{\text{hypotenuse}} = \frac{P}{S}$$

THREE-PHASE BALANCED CIRCUITS

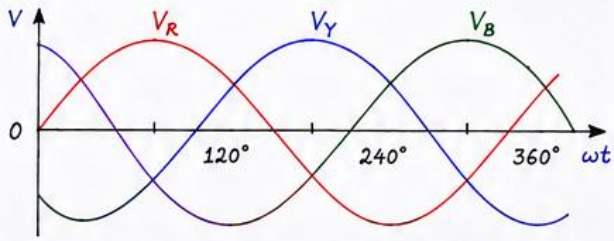
1. INTRODUCTION :

A three-phase system consists of three sinusoidal voltages (or currents) of equal magnitude and frequency, displaced in phase by 120° from each other. When the three phase quantities are equal in magnitude and phase displacement, the system is said to be balanced.

Advantages :

- Constant power to the load.
- More economical conductors (for same power).
- Smaller and lighter machines.
- Higher efficiency.

2. THREE-PHASE VOLTAGES :



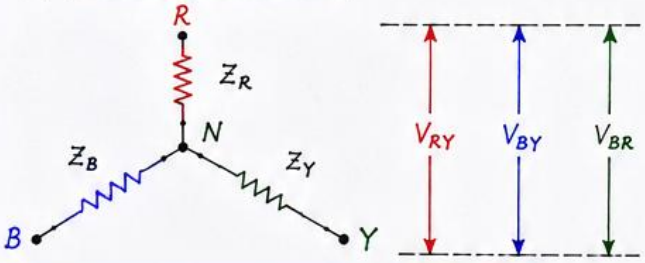
Instantaneous phase voltages :

$$V_R = V_m \sin \omega t$$
$$V_Y = V_m \sin (\omega t - 120°)$$
$$V_B = V_m \sin (\omega t - 240°)$$

V_m = Peak value (maximum)
V_{ph} = RMS value per phase
$\omega = 2\pi f$ (rad/s)
f = frequency (Hz)

3. CONNECTIONS OF THREE-PHASE SYSTEM

(A) STAR (Y) CONNECTION :



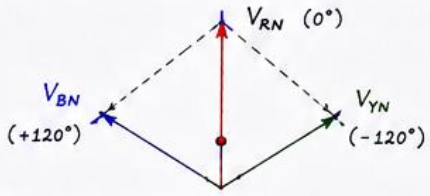
Line current = Phase current : $I_L = I_{ph}$

Line voltage = $\sqrt{3}$ × Phase voltage : $V_L = \sqrt{3}\, V_{ph}$

Derivation (Voltage relation) :

Consider phasor diagram of phase voltages.

V_{RN}, V_{YN}, V_{BN} are equal in magnitude and 120° apart.



Line voltage $V_{RY} = V_{RN} - V_{YN}$

$$|V_{RY}| = \sqrt{V_{RN}^2 + V_{YN}^2 - 2 V_{RN} V_{YN} \cos 120°}$$
$$= \sqrt{V_{ph}^2 + V_{ph}^2 - 2 V_{ph}^2 \left(-\tfrac{1}{2}\right)}$$
$$= \sqrt{3 V_{ph}^2} = \sqrt{3}\, V_{ph}$$

$V_L = \sqrt{3}\, V_{ph}$

Also, V_{RY} leads V_{RN} by 30°.

(Similarly for other line voltages)

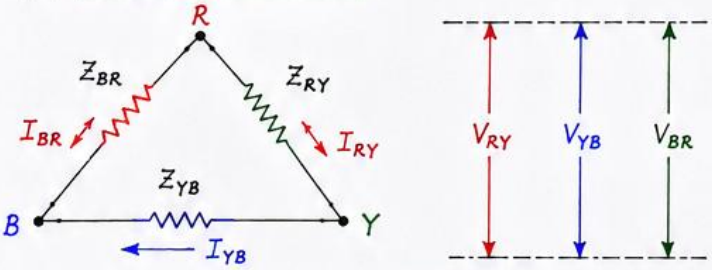
Line voltages :

$$V_{RY} = \sqrt{3}\, V_{ph} \angle 30°$$
$$V_{YB} = \sqrt{3}\, V_{ph} \angle 30°$$
$$V_{BR} = \sqrt{3}\, V_{ph} \angle 30°$$

Line currents :

$$I_R = I_{ph} \angle 0°$$
$$I_Y = I_{ph} \angle -120°$$
$$I_B = I_{ph} \angle +120°$$

(B) DELTA (Δ) CONNECTION :



Line voltage = Phase voltage : $V_L = V_{ph}$

Line current = $\sqrt{3}$ × Phase current : $I_L = \sqrt{3}\, I_{ph}$

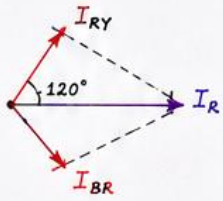
Derivation (Current relation) :

Consider R-junction. $I_R = I_{RY} - I_{BR}$

From phasor diagram of currents :

I_{RY} and I_{BR} are equal in magnitude I_{ph} and displaced by 120° (phase sequence R-Y-B).

Hence angle between I_{RY} and I_{BR} = 120°



$$|I_R| = \sqrt{I_{RY}^2 + I_{BR}^2 - 2 I_{RY} I_{BR} \cos 120°}$$
$$= \sqrt{I_{ph}^2 + I_{ph}^2 - 2 I_{ph}^2 \left(-\tfrac{1}{2}\right)}$$
$$= \sqrt{3 I_{ph}^2} = \sqrt{3}\, I_{ph}$$

$I_L = \sqrt{3}\, I_{ph}$

Phase currents :

$$I_{RY} = I_{ph} \angle 0°$$
$$I_{YB} = I_{ph} \angle -120°$$
$$I_{BR} = I_{ph} \angle +120°$$

Line currents :

$$I_R = \sqrt{3}\, I_{ph} \angle -30°$$
$$I_Y = \sqrt{3}\, I_{ph} \angle -150°$$
$$I_B = \sqrt{3}\, I_{ph} \angle +90°$$

NOTE :

- In a balanced system, the sum of the three phase voltages (or currents) at any instant is zero.
  $V_{RN} + V_{YN} + V_{BN} = 0$ and $I_R + I_Y + I_B = 0$
- The power delivered to a balanced three-phase load is constant (no pulsation).

POWER IN BALANCED THREE-PHASE CIRCUIT :

Total real power, $P = 3 V_{ph} I_{ph} \cos\phi = \sqrt{3}\, V_L I_L \cos\phi$ (Watt)

Total reactive power, $Q = 3 V_{ph} I_{ph} \sin\phi = \sqrt{3}\, V_L I_L \sin\phi$ (VAR)

Total apparent power, $S = 3 V_{ph} I_{ph} = \sqrt{3}\, V_L I_L$ (VA)

Power factor, $pf = \cos\phi = \frac{P}{S}$

Derivation of Relationship between Phase and Line Voltages in Star Connection using Vector Diagram

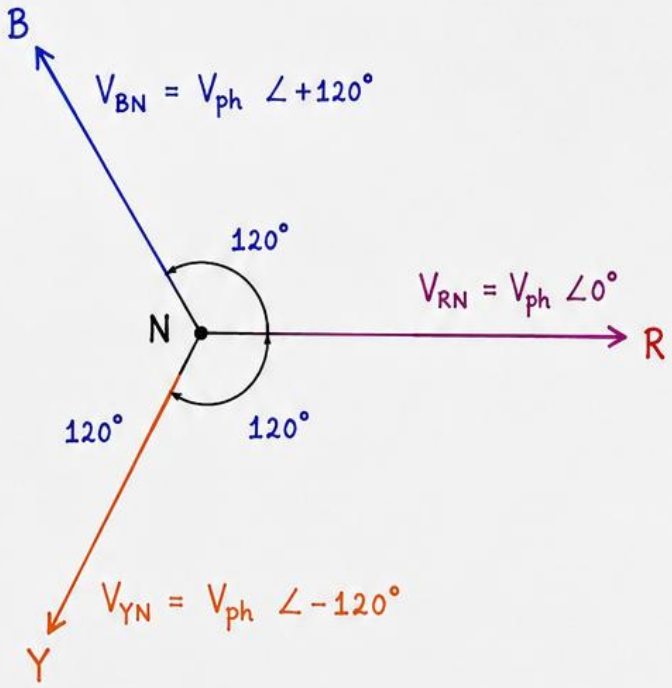
Consider a 3-phase balanced supply connected in star (Y) to a balanced load. Let the phase (line-to-neutral) voltages be :

$$V_{RN} = V_{ph} \angle 0°$$
$$V_{YN} = V_{ph} \angle -120°$$
$$V_{BN} = V_{ph} \angle +120°$$

The line (line-to-line) voltages are :

$$V_{RY} = V_{RN} - V_{YN}$$
$$V_{YB} = V_{YN} - V_{BN}$$
$$V_{BR} = V_{BN} - V_{RN}$$

1. Vector Diagram (Phase Voltages)



2. Line Voltages in terms of Phase Voltages

$$V_{RY} = V_{RN} - V_{YN}$$
$$= V_{ph} \angle 0° - V_{ph} \angle -120°$$
$$= V_{ph} \left[1 - (\cos -120° + j \sin -120°)\right]$$
$$= V_{ph} \left[1 - \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)\right]$$
$$= V_{ph} \left[\frac{3}{2} + j\frac{\sqrt{3}}{2}\right]$$
$$= V_{ph} \sqrt{3} \angle 30°$$

$$V_{YB} = V_{YN} - V_{BN}$$
$$= V_{ph} \angle -120° - V_{ph} \angle +120°$$
$$= V_{ph} \sqrt{3} \angle -90°$$

$$V_{BR} = V_{BN} - V_{RN}$$
$$= V_{ph} \angle +120° - V_{ph} \angle 0°$$
$$= V_{ph} \sqrt{3} \angle 150°$$

3. Magnitude Relationship

$$|V_{Line}| = \sqrt{3} |V_{Phase}|$$

i.e.. $V_L = \sqrt{3} V_{ph}$

4. Phase Relationship

- V_{RY} leads V_{RN} by 30°
- V_{YB} leads V_{YN} by 30°
- V_{BR} leads V_{BN} by 30°

Therefore, in star (Y) connection :

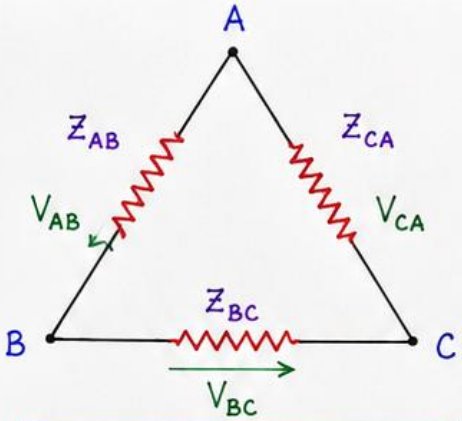
★ The line voltage is $\sqrt{3}$ times the phase voltage.

★ The line voltage leads the corresponding phase voltage by 30°.

Derivation of Relationship between Phase and Line Voltages in Delta Connection using Vector Diagram

1. Delta Connection

In a delta connected system, the three phases are connected in closed loop as shown in figure.



Phase (line to line) voltages :
V_{AB} , V_{BC} , V_{CA}

2. Let the phase (line to line) voltages be

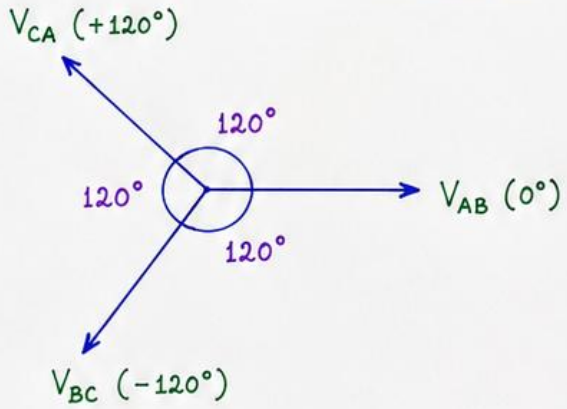
$$V_{AB} = V_{ph} \angle 0°$$
$$V_{BC} = V_{ph} \angle -120°$$
$$V_{CA} = V_{ph} \angle +120°$$

3. Line Voltages

Line voltage is the phasor difference between two phase voltages.

$$V_A = V_{AB} - V_{CA}$$
$$V_B = V_{BC} - V_{AB}$$
$$V_C = V_{CA} - V_{BC}$$

4. Vector Diagram of Phase Voltages



Hence,
$V_{AB} = V_{ph} \angle 0°$
$V_{BC} = V_{ph} \angle -120°$
$V_{CA} = V_{ph} \angle +120°$

5. Derivation (For line voltage V_A)

$$V_A = V_{AB} - V_{CA}$$

From vector diagram, V_{AB} and V_{CA} are 120° apart.

Using cosine rule,

$$|V_A|^2 = |V_{AB}|^2 + |V_{CA}|^2 - 2|V_{AB}||V_{CA}| \cos 120°$$
$$= V_{ph}^2 + V_{ph}^2 - 2 V_{ph} V_{ph} \left(-\tfrac{1}{2}\right)$$
$$= 2V_{ph}^2 + V_{ph}^2 = 3V_{ph}^2$$

$$|V_A| = \sqrt{3}\, V_{ph}$$

The angle of V_A with V_{AB} is 30° (leading).

Therefore,

$$V_A = \sqrt{3}\, V_{ph} \angle 30°$$

6. Results

* Magnitude of line voltage = $\sqrt{3}$ times the phase voltage.
* Line voltage leads its corresponding phase voltage by 30°.

Similarly,
$V_B = \sqrt{3}\, V_{ph} \angle -90°$
$V_C = \sqrt{3}\, V_{ph} \angle 150°$

Hence, in a balanced delta connection,
$V_L = \sqrt{3}\, V_{ph}$ and line voltage leads phase voltage by 30°.